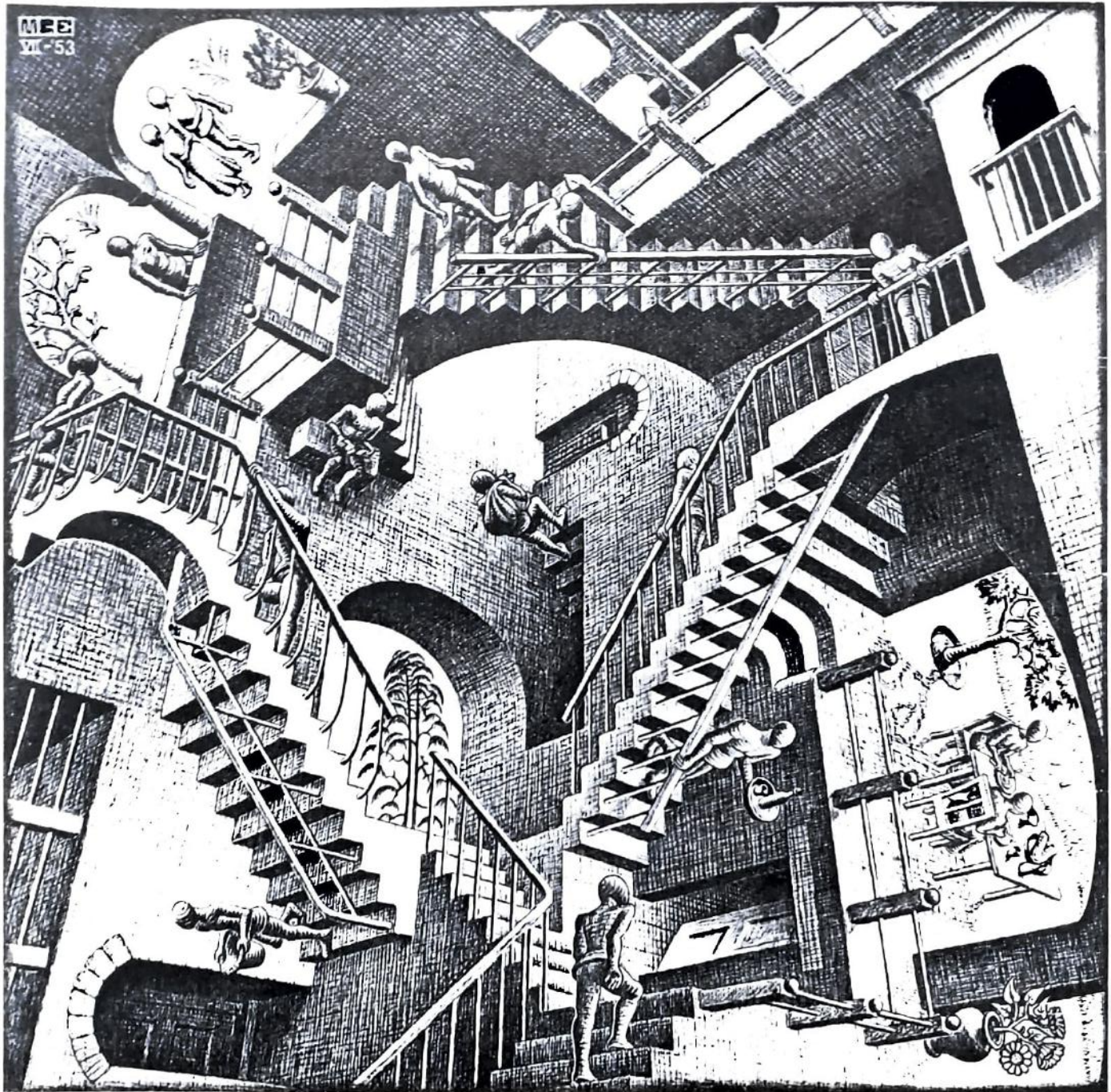


MATH SURVEY



STUYVESANT H.S. 1980

GENESIS*

Chapter 1

In the beginning Euclid created space.

2. And space was without structure, and void; and darkness was upon the face of deep space. And the spirit of Euclid moved.

3. And Euclid said let there be lines; and there were lines.

4. And Euclid saw the lines, that they were good; and Euclid divided the lines with points.

5. And Euclid called these lines rays. And the extent of the line was the first axiom.

6. And Euclid said let there be planes, and let them divide the points from the points.

7. And Euclid made planes and they divided the points which were under the planes from the points which were above the planes; and it was so.

8. And Euclid called the planes flat. And the evening and the morning were the second axiom.

9. And Euclid said let the points be gathered together unto one place, and let solids appear; and it was so.

10. And Euclid called the solids real; and the gathering together of the points he called space; and Euclid saw that it was good.

* This is a translation of an MS recently found in an ancient wine vessel in a cave in the mountains of Iraq. There are missing sections because of the fragmentary nature of the MS.

11. And Euclid said, let the space bring forth segments, the segments yielding fruit after their kind whose seed is in itself; and it was so.

12. And space brought forth segments, yielding segments upon segments to make number lines; and Euclid saw that it was good.

13. And the evening and the morning were the third axiom.

14. And Euclid said let there be orbs in the firmament of space to divide the inside from the outside and let them be signs for the seasons, and for days, and for years.

15. And let them be for gems in the firmament of space to give delight upon the earth; and it was so.

16, 17, 18. And Euclid made two great orbs the greater orb to rule the firmament and the lesser orb to rule the planes; he made the triangles also; and Euclid saw that it was good.

19. And the evening and the morning were the fourth axiom.

20, 21. And Euclid said let the points bring forth abundantly parallels so that alternate interiors are forever equal. And Euclid created great cubes, and squares, and every parallelogram that cometh, which the points brought forth abundantly, after their kind one upon another; and Euclid saw that it was good.

22. And Euclid blessed them, saying, be fruitful and multiply and with parallelograms and parallelepipeds fill the planes and space.

23. And the evening and the morning were the fifth axiom.

26. And Euclid said let us make a Mathematician in our image, after our likeness; and let him have dominion over triangles and over circles, and over spheres and over cubes, and over all of space, and every constructable subset that is constructed out of space.

27. So Euclid created a Mathematician in his own image, in the image of Euclid created he him, great and small created he them.

28. And Euclid blessed them, and Euclid said unto them, be fruitful, and multiply and prove theorems and subdue them; and have dominion over triangles, and over spheres, and over every set of points that cometh out of space.

29. And Euclid said Behold, I have given you every set bearing proof, which is on the face of all of space, and every set which is the fruit of a set yielding proof; to you it shall be for theorems.

31. And Euclid saw everything that he had made, and, behold, it was very good. And the evening and the morning were the sixth day.

ON THE COVER

The cover is a print entitled "Relativity" by M.C. Escher. Escher, a Dutch artist born in 1898, has said that he feels closer to mathematicians than to his fellow artists.

"Relativity" portrays a three dimensional world with gravitational forces in three different directions. The people living in each gravitational field are unaware of the other worlds which coincide with their own. The picture is symmetric and the worlds are identical, or are they?

MATH SURVEY

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How Far Can You See?

by Zachary Franco

Little Jimmy finally landed on Earth- plop!- right in the middle of the Sahara Desert. He had been invited to Earth by Professor 'Bud' Riemann, but has apparently mixed up his directions (you will excuse him, I'm sure-he just returned from Null Space after rescuing Professor Euclidean from the 0th dimension).

Little Jimmy was approached by three sheiks, who, upon seeing his crown studded with seventeen diamonds, offered him all the land around him, as far he could see in exchange for it. Little Jimmy, who was no fool, wanted to know how much land he was getting, so he pulled out his pocket telephone and dialed 164-4934:

"Professor Riemann, what is the surface area of a spherical sector formed by the locus of the points of tangency of tangents drawn from a point 5 feet above a sphere, to the sphere, if its radius is 21 million feet?"

"You should be ashamed of yourself, Jim! If you'd just plug all that into your GAUSS-17 pocket calculator, you would find the area to be 21.665 square miles, or 15,146 acres," the professor shrieked.

"That means that I would receive 15,146 acres, each worth \$4 (desert is cheap in Africa), a total of \$60,584, whereas each diamond in my crown is worth \$3500, and the band is worth \$1085, so my crown is worth \$60,585," calculated Jimmy. "Wow! If I hadn't known that, Bud, I might have sold my crown and lost a dollar," remarked Jimmy. The three sulking sheiks left and have since become Idi Amin Dada, Colonel Qadhafi, and the Ayatollah Khomeini.

"But how did you know the answer so quickly?" resumed Jimmy.

"Because I am a genius," Professor Riemann modestly replied.

"But that's what you always say- can't you ever explain your methods?"

"Alright," replied Professor Riemann, "but just this one time- for Stuyvesant's 1980 Math Survey. Take a circle and a point P outside it (see diagram 1). Using the Pythagorean Theorem,

$$OP^2 = OB^2 + BP^2, \text{ or } (r+h)^2 = t^2 + r^2,$$
 from which we solve for t :
$$t = \sqrt{h^2 + 2rh}.$$
 This means that if h is the distance from your eye to the ground, in feet, and the radius of the earth is r feet, then the distance from your eye to the horizon is $\sqrt{h^2 + 2rh}$ feet, or $(\sqrt{h^2 + 2rh})/5280$ miles. Since h is small, compared to r , we may approximate this as $(\sqrt{2r/5280})\sqrt{h} \approx \frac{11}{9}\sqrt{h}$. If you are picky and want the curved distance from foot to horizon, it is $r \cdot \text{Arctan}(t/r)$."

"But how did you find the surface area, the amount of land that one can see?", persisted Jimmy.

"Calm down! I am getting to that. In diagram 2, C is the foot of the altitude from B to PO, and s is the distance from C to the

surface. Triangles PCB and PBO are similar, so $PC/t = t/PO$, or $PC = t^2/(r+h)$ and $s = PC - h = t^2/(r+h) - h = (h^2 + 2rh)/(r+h) - h = rh/(r+h)$.

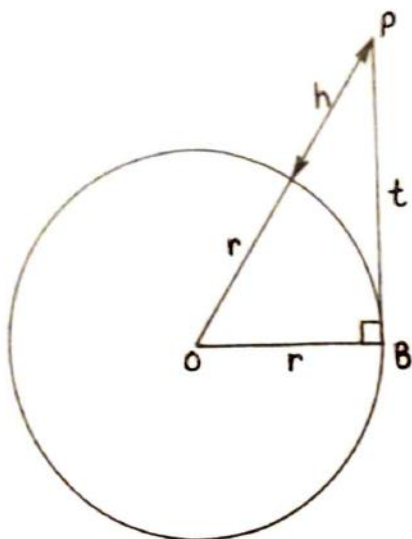


Diagram 1

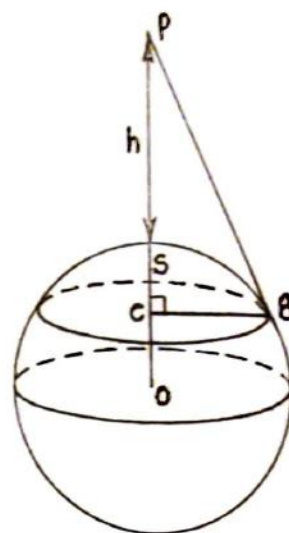


Diagram 2

The surface area of the shaded sector $= 2\pi rs = 2\pi r^2 h/(r+h)$. A proof of this can be found in most solid geometry texts. Since h is small compared to r , the area $\approx 2\pi r^2 h/r \approx 4.73h$ square miles, where h is in feet. So if you are 5 feet tall, you can see about $4.73(5)$, or 23.65 square miles of land, whereas a six-footer can see about 28.38 square miles of land. Note that in this solution, we take $r = 21$ million feet."

How far can you see?

We assume that the distance from your eyes to the top of your head is 4". If you have an unusually large or small forehead, make appropriate adjustments.

<u>your height (h+4)</u>	<u>how far can you see ($11\sqrt{h/9}$)</u>
4' 8" - 5' 0"	2.6 miles
5' 1" - 5' 4"	2.7 miles
5' 5" - 5' 9"	2.8 miles
5' 10" - 6' 1"	2.9 miles
10 feet	3.8 miles
15 feet	4.7 miles
20 feet	5.4 miles
25 feet	6.1 miles
50 feet	8.6 miles
100 feet	12.2 miles
500 feet	27.3 miles
1000 feet	38.6 miles

The actual values are slightly smaller due to air conditions and the varying radii of the earth. However, these results do check with the Weather Bureau's Visibility Figures to within 5%. Note that for larger values of h , the error becomes larger.

Many Primes

by Cathy Savage

There are many different ways to prove the existence of infinitely many primes. Shown below are four such proofs.

Proof 1: A Fermat number is an integer of the form $F_k = 2^{2^k} + 1$.

Lemma: $F_0 F_1 F_2 \dots F_k = F_{k+1} - 2$.

Proof: For $k=0$, direct substitution yields $3=3$.

Proceeding by induction,

$$\begin{aligned} (F_0 F_1 F_2 \dots F_k) F_{k+1} &= (F_{k+1} - 2) F_{k+1} \\ &= (2^{2^{k+1}} - 1)(2^{2^{k+1}} + 1) \\ &= (2^{2^{k+1}})^2 - 1 \\ &= 2^{2^{k+2}} - 1 \end{aligned}$$

$= F_{k+2} - 2$, thus establishing the Lemma.

Assume that some prime p divides both F_m and F_n (with $m < n$). Since $p | F_n$ and $p | F_0 F_1 \dots F_{n-1}$, it follows that

$$p | (F_n - F_0 F_1 \dots F_{n-1}) = 2.$$

Since the only prime factor of 2 is 2 itself, we must have $p=2$.

But $2^{2^m} + 1$ is odd, so it cannot possibly be a multiple of p , contradicting our assumption. Since there are infinitely many Fermat numbers, it follows that there are infinitely many primes.

Proof 2: We shall make use of the following Lemmas:

Lemma 1: If p is a prime number, then $n!$ has exactly $\left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + \dots$ factors of p (Where $[x]$ denotes the largest integer not exceeding x).

Proof: The expression $[n/p]$ counts the number of positive integers less than or equal to n which are multiples of p , each of which contributes one factor of p to $n!$; $[n/p^2]$ counts the number of integers not exceeding n which are multiples of p^2 , each of which contributes an additional factor of p to $n!$; etc. By extending our sum indefinitely, every factor of p in $n!$ will be counted exactly once.

Lemma 2: $\sqrt[n]{n!} \geq \sqrt[2]{n}$.

Proof: $(n!)^2 = \binom{n}{1} \binom{n}{2} \binom{n}{3} \dots \binom{n}{n}$
 $\geq (n) \cdot (n) \cdot (n) \cdot \dots \cdot (n)$ (n terms)
 $= n^n$.

Taking the $(2n)$ th root of both sides, we obtain the desired result.

Let $p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$ the prime factorization of $n!$.

Then $a_i = \left\lfloor \frac{n}{p_i} \right\rfloor + \left\lfloor \frac{n}{p_i^2} \right\rfloor + \left\lfloor \frac{n}{p_i^3} \right\rfloor + \dots$ (By Lemma 1)

$$\leq \frac{n}{p_i} + \frac{n}{p_i^2} + \frac{n}{p_i^3} + \dots \quad (\text{Since } [x] \leq x)$$

$$= \frac{n/p_i}{1 - 1/p_i} \quad (\text{Summing a geometric series})$$

$$= \frac{n}{p_i - 1},$$

and therefore

$$n! \leq p_1^{n/(p_1-1)} p_2^{n/(p_2-1)} p_3^{n/(p_3-1)} \dots p_k^{n/(p_k-1)}.$$

Taking the n th root of both sides, we have

$$\sqrt[n]{n!} \leq p_1^{1/(p_1-1)} p_2^{1/(p_2-1)} p_3^{1/(p_3-1)} \dots p_k^{1/(p_k-1)}.$$

By Lemma 2, $\sqrt[n]{n!} \geq \sqrt[2]{n}$, hence,

$$n \leq p_1^{1/(p_1-1)} p_2^{1/(p_2-1)} p_3^{1/(p_3-1)} \dots p_k^{1/(p_k-1)}.$$

But n can be as large as we please, contradicting the fact that it is always less than the finite quantity on the right side of the last inequality. Therefore there are infinitely many primes.

Proof 3: Assume there is a finite number of primes; call them $p_1, p_2, \dots, p_n$

Consider the product $\prod_{i=1}^n \frac{1}{1 - (1/p_i)}$.

Expressing each term as the sum of a geometric series, we have

$$\prod_{i=1}^n \frac{1}{1 - (1/p_i)} = \prod_{i=1}^n \left(1 + \frac{1}{p_i} + \frac{1}{p_i^2} + \dots \right)$$

$$= \left(1 + \frac{1}{p_1} + \frac{1}{p_1^2} + \dots \right) \left(1 + \frac{1}{p_2} + \frac{1}{p_2^2} + \dots \right) \dots \left(1 + \frac{1}{p_n} + \frac{1}{p_n^2} + \dots \right)$$

The reciprocal of every prime to every power appears somewhere in this expression. By choosing the appropriate term from each of the n factors, we can obtain the reciprocal of the prime factorization of any integer. Therefore,

cont. on page 31

A Problem in Combinatorics

by Alan Horowitz

$\binom{n}{k}$ (pronounced n choose k) is the number of ways k objects can be chosen from n objects. For example, $\binom{52}{5}$ represents the number of five card poker hands in a standard deck. It can be shown that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Suppose we have a deck of n distinguishable cards. It is tossed in the air and the cards that land face-up (if any) are collected. How many different combinations of face-up cards can occur, given that the number of cards landing face-up is divisible by three? In other words, we wish to find the sum S of the number of possible outcomes when $0, 3, 6, 9, \dots$ cards land face-up.

$$\text{Thus } S = \binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \binom{n}{9} + \dots$$

To help evaluate this let $w = \cos 120^\circ + i \sin 120^\circ = (-1 + \sqrt{3}i)/2$. By DeMoivre's Theorem, $w^{3k} = \cos(360k^\circ) + i \sin(360k^\circ) = 1 + 0i = 1$,
 $w^{3k+1} = \cos(360k+120)^\circ + i \sin(360k+120)^\circ = w$,
 and $w^{3k+2} = \cos(360k+240)^\circ + i \sin(360k+240)^\circ = w^2$.
 Recall the binomial theorem: $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$. Applying this,

$$(1) \quad (1+1)^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \binom{n}{4} + \dots$$

$$(2) \quad (1+w)^n = \binom{n}{0} + \binom{n}{1}w + \binom{n}{2}w^2 + \binom{n}{3}(1) + \binom{n}{4}w + \dots$$

$$(3) \quad (1+w^2)^n = \binom{n}{0} + \binom{n}{1}w^2 + \binom{n}{2}w + \binom{n}{3}(1) + \binom{n}{4}w^2 + \dots$$

Summing (1), (2), and (3)

$$2^n + (1+w)^n + (1+w^2)^n = 3\binom{n}{0} + (1+w+w^2)\binom{n}{1} + (1+w+w^2)\binom{n}{2} + \dots$$

Since $w^3 - 1 = 0$, $(w-1)(w^2+w+1) = 0$, but $w \neq 1$, so $w^2+w+1 = 0$.

$$\text{Therefore, } 2^n + (1+w)^n + (1+w^2)^n = 3\binom{n}{0} + 3\binom{n}{3} + 3\binom{n}{6} + \dots$$

$$\text{and } S = (2^n + (1+w)^n + (1+w^2)^n)/3$$

$$\text{Since } w = (-1 + \sqrt{3}i)/2, \quad w+1 = (1 + \sqrt{3}i)/2 = \cos 60^\circ + i \sin 60^\circ, \\ w^2 = (-1 - \sqrt{3}i)/2, \quad w^2+1 = (1 - \sqrt{3}i)/2 = \cos(-60^\circ) + i \sin(-60^\circ),$$

$$S = (2^n + \cos(60n) + i \sin(60n) + \cos(-60n) + i \sin(-60n))/3 \\ = (2^n + 2\cos(60n))/3.$$

Therefore, there are $(2^n + 2\cos(60n))/3$ possible combinations of the face-up cards.

In the general case of the series $\binom{n}{a} + \binom{n}{2a} + \binom{n}{3a} + \dots$, a similar procedure to the one above yields:

$$\left(\sum_{k=1}^n (1+w^k)^n \right) / a, \text{ where } w = \cos(360/a)^\circ +$$

$i \sin(360/a)^\circ$. However, for larger a , $(1+w)^n$ is not easily evaluated by De Moivre's Theorem.

Math Society

Math Society brings to Stuyvesant students lectures about various mathematical topics. The atmosphere is a relaxed one, where students are encouraged to pose questions and mention other thoughts. These ideas can be discussed after the lecture over cookies, soda, and sometimes peanuts. Because of the friendly atmosphere, record numbers of students have shown up after school this year. We hope to have even more in the future.

At a Math Society lecture one might be introduced to any of the following topics: Game Theory, "all about e ", probabilities of horse racing, Monte Carlo Methods, and Non Euclidean Geometry. Topics ordinarily not introduced until college, are presented on a level which any interested student can understand. Our lecturers this year have been superb. Among them have been: Peter Lax, director of the Courant Institute of N.Y.U., David C. Kelly, of Hampshire College fame; and Haig Bohigian of John Jay College. In past years we have seen such films as; "Space Filling Curves" and "Donald Duck in Mathamagic land." These films have always been highly successful.

This year we have been blessed with the dynamic leadership of Steven Lev, who unfortunately is moving on to bigger and better things. This leaves room for people who are willing to work. If you have any ideas for future programs or wish to help with publicity, refreshments, or contacting speakers, come to one of our meetings or drop by at Math Team any day before first period. Try a new angle in life (mathematically and socially). Join us!

Burnside's Lemma

by Lisa Randall

Suppose we were asked to find the number of ways in which one could color the faces of a tetrahedron with k colors so that no colorings are the same. One might first conjecture 4^k ways, since each face can be colored with k different colors. But we find that many colorings are actually the same, because of the symmetry of the tetrahedron. In order to solve this problem and many similar ones, a useful technique is Burnside's Lemma.

Let us consider a group, G , acting on a set, S . Our set will consist of permutations of this set (for example, the various rotations of a tetrahedron). It is easy to show that there is an identity and inverse for each element, closure, and associativity, so we see that the set of permutations is indeed a group. In general, there are $m!$ permutations of m elements. However, in the problems in which the lemma is employed, we will generally only need a subgroup of the set of all permutations on m elements. For example, when we consider a rigid body such as a tetrahedron, adjacent sides remain adjacent so that not all permutations of the faces are possible.

Next, we consider the orbit defined on a subgroup of G , where the orbit of an element is the set of all its images under the subgroup. An orbit may be considered a subset which is mapped into itself. There are usually more than one orbit, since not all elements are mapped into each other. We will show the number of different orbits = $\frac{1}{|G|} \cdot \sum_{p \in G} T(p)$ (1) where $T(p)$ is the number of orbits fixed by p . As an example, consider what happens when we flip a square across the line connecting the midpoints of two opposite edges. The top and bottom edge will remain fixed and can be colored any color. The two vertical edges must be colored the same color since one will be mapped into the other. We could have three independent colors, or orbits in this example.

Considering equation (1), we find it is equivalent to the number of orbits = $\frac{1}{|G|} \sum_{s \in S} z(s)$ (2) where $z(s)$ = the number of permutations that fix s and S is the set of permutations, since in both cases we are adding the number of times all the elements are fixed. We will find the second easier to prove, while the first is better for calculations.

Let us consider an arbitrary element of our set, a . Then the number of elements which fix a is $z(a)$. We see that the number of permutations which take a to b is at least $z(a)$, since if $g(a)=b$ and $h(a)=a$, clearly $g(h(a))=b$. This holds for all h which fix a . But we see that if $g(h_1(a))=g(h_2(a))$, multiplying by g^{-1} , we get $h_1(a)=h_2(a)$. Therefore, each operation is unique so that there are exactly $z(a)$ permutations which take a to b . Similarly, there

will be exactly $z(a)$ permutations which take a to c , etc., so that

$$z(a) = \frac{|G|}{\text{size of orbit}}, \text{ where } |G| \text{ is the order of our group.}$$

It can be shown that orbits are either disjoint or identical, although not all orbits are necessarily the same size. Therefore, they will partition our set into disjoint equivalence classes.

We know $\sum_{s \in G} z(s) = |G|$.

Since the distinct orbits are disjoint, we know

$$\sum_{\text{all orbits}} \sum_{s \in \text{particular orbit}} z(s) = |G| \cdot (\text{the number of orbits})$$

Therefore, if we sum over all $s \in S$, we get

$$\sum_{s \in S} z(s) = |G| \cdot (\text{the number of orbits})$$

or, Burnside's Lemma,

$$\text{the number of different orbits} = \frac{1}{|G|} \sum_{p \in G} T(p).$$

Now, let us apply the lemma to our original example, coloring a tetrahedron with k colors. We need to find the symmetries of a tetrahedron and find the orbit for each.

Let p_1 be the identity. All faces are fixed so $T(p_1) = k^6$, since each face may be colored independently.

Let p_2, p_3, p_4, p_5 be the operations which fix one vertex and rotate the opposite face 120° . Since there are four vertices, there are four such operations. Then $T(p_2) = T(p_3) = T(p_4) = T(p_5) = k^2$, since there are two orbits.

Let p_6, p_7, p_8, p_9 be the operations which fix one vertex and rotate the base triangle 240° . Then $T(p_6) = T(p_7) = T(p_8) = T(p_9) = k^2$.

Let p_{10}, p_{11}, p_{12} be the 180° rotation about the axis which connects the midpoint of two opposite edges. Then $T(p_{10}) = T(p_{11}) = T(p_{12}) = k^2$.

Applying Burnside's Lemma, we get

$$\text{the number of ways to color a tetrahedron with } k \text{ colors} = \frac{1}{12} (k^4 + 8k^2 + 3k^2).$$

For example, the number of ways to color the faces of a tetrahedron with 2 colors is 5.

COMPLIMENTS OF THE

PARENT'S ASSOCIATION

The Prime Number Theorem

by Daniel Glasser

The sequence of prime numbers 2, 3, 5, 7, 11, 13, 17.. has intrigued mathematicians for thousands of years. It is known that there are infinitely many primes (see page), but their occurrence follows no specific pattern, and the chance of finding one is slim. Knowing this, one might think it impossible to set reasonable bounds on the number of primes less than a given integer. (One could say that for sufficiently large n , the number of primes less than or equal to n is between $2/n$ and $n/2$, but these bounds become so far apart that they tell us nothing about the behavior of the function.) Surprisingly, this is false. The Russian mathematician Chebyshev proved in 1850 that for sufficiently large n , the number of primes less than or equal to n is within 11% of $n/\ln n$. If we define $F(n)$ as the number of primes less than or equal to n (e.g. $F(1)=0$, $F(3)=2$, $F(10)=4$), this result may be expressed mathematically as:

$$\frac{.89n}{\ln(n)} < F(n) < \frac{1.11n}{\ln(n)}.$$

We will not prove this, but we will show that:

$$\frac{2n}{3\ln(n)} < F(n) < \frac{1.7n}{\ln(n)}.$$

To prove the upper bound, we will make the inductive assumption that it holds for all integers $n < k$.

Case 1: $k=2q$.

Consider the binomial coefficient

$$\binom{2q}{q} = \frac{(2q)(2q-1)(2q-2)\dots(q+1)}{(q)(q-1)(q-2)\dots(1)}.$$

Since every prime between q and $2q$ appears in the numerator but not in the denominator, $\binom{2q}{q}$ is divisible by every prime between q

and $2q$. There are $F(2q) - F(q)$ of these primes, each one greater than q . Therefore

$$F(2q) - F(q) < \frac{\binom{2q}{q}}{q}.$$

But

$$\binom{2q}{q} < \binom{2q}{1} + \binom{2q}{2} + \dots + \binom{2q}{q} + \dots + \binom{2q}{2q} = 2^{2q}, \text{ so}$$

$$\frac{F(2q) - F(q)}{q} < \frac{2^{2q}}{2}.$$

Taking the natural logarithm of both sides,

$$(F(2q) - F(q)) \ln q < 2q(\ln 2),$$

$$F(2q) - F(q) < 2q(\ln 2) / \ln q,$$

$$F(2q) < F(q) + 1.39q / \ln q.$$

Since we know by our inductive hypothesis that $F(q) < 1.7q / \ln q$, we get

$$F(2q) < 1.39q / \ln q + 1.7q / \ln q = 3.09q / \ln q.$$

Using either calculus or a calculator, it can be shown that $3.09q / \ln q < 1.7(2q) / \ln(2q)$ (for $q > 995$), and hence

$$F(2q) < \frac{1.7(2q)}{\ln(2q)}.$$

Case 2: $k=2q+1$.

Since $F(2q+1) \leq 1+F(2q)$, our hypothesis implies that $F(2q+1) < 1+(3.09q/\ln q)$.

Again, using either of the two methods mentioned above, one may show that

$$1+(3.09q/\ln q) < 1.7(2q+1)/\ln(2q+1) \text{ (for } q > 1123), \text{ and so}$$

$$F(2q+1) < \frac{1.7(2q+1)}{\ln(2q+1)}.$$

Since we have shown that truth of the Lemma for $n \leq k$ implies truth for k , whether k is odd or even, all that is needed to complete the induction are the appropriate base cases. Since the above inequalities hold for $q > 1123$, we need to test the upper bound for all integers less than 1123, which is a simple task for a calculator or a computer. (Incidentally, it does check out.)

For the lower bound, we do not need induction, but we do need the following:

Lemma: If x is the greatest power of p (where p is a prime number) dividing $\binom{n}{r}$, then $p^x \leq n$.

Proof: One can show that the greatest power of p dividing $a!$ is $\left[\frac{a}{p}\right] + \left[\frac{a}{p^2}\right] + \left[\frac{a}{p^3}\right] + \left[\frac{a}{p^4}\right] + \dots$, where $[k]$ denotes the greatest integer not exceeding k . Using this result, we find that

$$x = \sum_{k=1}^{\infty} \left\{ \left[\frac{n}{p^k} \right] - \left[\frac{r}{p^k} \right] - \left[\frac{n-r}{p^k} \right] \right\}.$$

Each term of the summation is either 0 or 1, and is zero whenever $p^k > n$ (or equivalently, when $k > \ln n / \ln p$), which implies that there are at most $\ln(n)/\ln(p)$ terms whose value is 1. This means that $x \leq \ln(n)/\ln(p)$, or that $p^x \leq n$.

Back to the main proof:

By the Fundamental Theorem of Arithmetic, $\binom{n}{r}$ may be expressed as $(p_1^{a_1})(p_2^{a_2})(p_3^{a_3})\dots(p_k^{a_k})$, where the p_i are distinct prime numbers.

Since $\binom{n}{r}$ is not divisible by any prime greater than n , $k \leq F(n)$.

According to the Lemma, each $p_i^{a_i}$ is less than or equal to n . Therefore $\binom{n}{r} \leq n^k \leq n^{F(n)}$.

Since $2^n = \sum_{r=0}^n \binom{n}{r} \leq \sum_{r=0}^n n^{F(n)} = (n+1)n^{F(n)}$,

it follows (by taking logarithms) that

$$n(\ln 2) \leq \ln(n+1) + F(n)(\ln n),$$

or

$$\frac{n(\ln 2) - \ln(n+1)}{\ln(n)} \leq F(n).$$

For the third time, we may use calculus or a calculator to verify that $\frac{2n}{3\ln(n)} < \frac{n(\ln 2) - \ln(n+1)}{\ln(n)}$ (for $n > 198$), which implies that

$\frac{2n}{3\ln(n)} < F(n)$. Using computer or calculator, the inequality can be verified for $n < 198$, and our theorem is proven.

An Introduction to Logic

by Leonid Fridman

One definition of mathematical logic is the analysis of mathematical proof. Until about 90 years ago, logic was not considered to be an important discipline. Its apparently self-evident character discouraged any deep logical investigations. Then, around the turn of the century, the mathematical world was shocked by the discovery of the paradoxes, i.e. arguments leading to contradictions.

One of the most famous and understandable paradoxes is Russell's paradox (1902):

Recall that by a set we mean any collection of objects, called elements of the set. Some examples are the set of all integers, the set of all cats in New Jersey, the set of all sets. Call a set "usual" if it does not contain itself as an element. For example the set of all integers is "usual", while the set of all sets is not. From definition, every set is either "usual" or not "usual". Now consider the set A of all "usual" sets. Is A "usual" or not?

Case I Suppose A is "usual". Then it must contain itself, by definition of A . But every set that contains itself is not "usual". Thus we have a contradiction.

Case II Suppose the set A is not "usual". Then it cannot contain itself, by definition of A . But then, by definition of "usual", A is "usual". Another contradiction.

Thus we have shown that A is both "usual" and not "usual".

A series of "proofs" like this forced mathematicians to reexamine their notions of sets and proofs. One result of these efforts was the foundation of rigorous set theory based on Zermelo-Skolem-Fraenkel axioms. Another was the great interest generated in mathematical logic, which, together with set theory, became the foundation of modern mathematics. The results of Gödel, Tarski, Church and many others were very deep and devastating. In 1931 a German mathematician, K. Gödel, proved one of the most important and beautiful theorems in mathematics—Gödel's Incompleteness Theorem.

The theorem, very roughly, states that in every axiomatic system Γ (such as the theory of natural numbers or Euclidean Geometry) there exists a statement A , expressed in the language of the system, such that A is not provable or disprovable in Γ . For example, Kahn in 1963 showed that the Continuum Hypothesis was such a statement in set theory. And many mathematicians believe that Fermat's Last Theorem (which states that for every $n \geq 3$ there do not exist natural numbers a, b, c , such that $a^n + b^n = c^n$) is such a statement in number theory. If that is true, that would mean that no matter how long mathematicians would work on this problem, and no matter how smart they were, they would never be able to prove it, nor disprove it.

The Gödel's Incompleteness Theorem can easily be extended to show that for every axiomatic system Γ there exist infinitely many statements $A_1, A_2, \dots, A_n, \dots$ which cannot be proven or disproven in Γ . Gödel's theorem already guarantees us one of those statements A . Now make A a new axiom of the system (keeping

all the others). The new system $(\Gamma + A_1)$ will still be consistent since Γ tells us nothing about A_1 . Now apply Gödel's theorem again, which assures that there exists a statement A_2 , which is undecidable from Γ and A_1 . Thus it is clearly not provable or disprovable from Γ alone. If we continue this process, we will see that there have to be infinitely many statements $A_1, A_2, \dots, A_n, \dots$ which are undecidable in Γ .

Westinghouse Tips

For the past thirty-nine years, Westinghouse Corporation has sponsored a special kind of competition. Students throughout the nation submit reports on scientific research they have conducted, either individually or under the guidance of a teacher, professor or some other knowledgeable person. About a thousand papers are received annually, of which three hundred are designated Honorable Mention. Of these, forty are picked to be winners. The winners, who receive an all-expenses paid trip to Washington, D.C., get the chance to compete for ten scholarships ranging from \$5,000 to the top prize of \$12,000. Winners not receiving scholarships all get \$500 cash awards.

Most of the papers received deal with either experiments in biology, problems in mathematics, or the physical aspects of some engineering concern. Access to giant laboratories, nuclear reactors, electron microscopes and particle accelerators should not be considered as essential, although facilities superior to the Stuyvesant High School labs are desirable for most projects not in the field of math. Since this is a math magazine, suggestions will be made on writing math papers (which have had extraordinary success, especially among Stuyvesant students), some of which may apply to other types of projects as well.

Unlike most local math or science fairs, the judges who grade the papers are experts in their fields. The chances are good that they have a background in your particular field of study, thus you may write on a sophisticated level without explaining every detail of a proof. However, the judges will be aware of holes in a proof (such as cases not considered), hand-waving, faked results and the famous fudging techniques. For these reasons, it is a good, sound idea to make proofs as rigorous as possible. Also, label all your theorems, lemmas, postulates, or conjectures. Try to emphasize as much as possible the significance of what you have done.

Finding a topic on which to do a project is one of the hardest parts. Unsolved problems are nice, although if they are well known, it is unlikely that you will prove anything new. It is in most cases preferable to invent your own problem, perhaps looking

(cont. on page 18)

Ménages Problem

by Gregg Patruno

Recent historical discoveries indicate that King Arthur ran a computer dating service for the Knights of the Round Table as early as 517 AD. (My unquestionably reliable source wishes to remain nameless for some reason.) Picture n Knights alternating with n Ladies around the table, anxiously awaiting the printout. What is the probability that none of the soon-to-be-announced couples was sitting together?



(We assume throughout that the Knights took their seats randomly after the Ladies had been seated in the even-numbered chairs.)

We shall need the following Lemma:

If $P(i_1, i_2, \dots, i_m)$ represents the number of seating arrangements for which the Ladies in seats $2i_1, 2i_2, \dots, 2i_m$ all sit next to their own Knights, then the number of seating arrangements for which no couple sits together is

$$S = n! - \sum_{i=1}^n P(i_1) + \sum_{1 \leq i_1 < i_2 \leq n} P(i_1, i_2) - \sum_{1 \leq i_1 < i_2 < i_3 \leq n} P(i_1, i_2, i_3) + \dots + (-1)^n P(1, 2, 3, \dots, n).$$

(Such arrangements will be called "acceptable".)

Proof:

Suppose that a certain seating arrangement has exactly r couples sitting together, $r \geq 1$. Since $n!$ counts all possible permutations of n Knights in n chairs, this arrangement will be counted exactly once by the $n!$ term in S . For each $k \leq r$, the $\sum P(i_1, i_2, \dots, i_k)$ term counts the arrangement exactly $\binom{r}{k}$ times, corresponding to each way of choosing k of the r couples. The arrangement is not counted at all by $\sum P(i_1, i_2, \dots, i_k)$ if $k > r$, because we would not have k couples seated together. Therefore, the arrangement is counted a total of

$$1 - \binom{r}{1} + \binom{r}{2} - \binom{r}{3} + \dots + (-1)^r \binom{r}{r} + 0 + 0 + \dots$$

times by S . But this is the binomial expansion of $(1+(-1))^r$, and consequently equals zero. Thus, any arrangement with couples seated together is not counted by S .

Every acceptable arrangement is counted once by the $n!$ term and is not counted by any of the succeeding terms, hence S is precisely the number of acceptable seating arrangements.

Q.E.D.

We shall now find an expression for $\sum P(i_1, i_2, \dots, i_k)$, which may be interpreted as the number of ways of first seating k couples together and then filling the remaining seats.

Case 1: The Knight in seat 1 and the Lady in seat $2n$ are not one of the k couples.

One of King Arthur's attendants brings out k loveseats for the lucky couples and $2n-2k$ armchairs for the other guests, and places them around the table starting clockwise from seat 1. Since he must choose k positions for the loveseats in the sequence of $2n-k$ pieces of furniture, there are $\binom{2n-k}{k}$ ways he can do so. After the k lucky Knights are seated next to their Ladies, the $n-k$ unattached Knights will have $n-k$ seats to choose from, which they may fill in any of $(n-k)!$ ways, yielding a total of $\binom{2n-k}{k}(n-k)!$ arrangements.

Case 2: The Knight in seat 1 and the Lady in seat $2n$ are one of the k couples.

One of the loveseats must be placed in positions 1 and $2n$. The attendant then chooses the positions for the $k-1$ other loveseats among the $2n-k-1$ remaining pieces of furniture; there are $\binom{2n-k-1}{k-1}$ such ways. The $n-k$ unattached Knights again have $(n-k)!$ ways to take their seats, for a total of $\binom{2n-k-1}{k-1}(n-k)!$ arrangements.

Hence

$$\begin{aligned} \sum P(i_1, i_2, \dots, i_k) &= \binom{2n-k}{k}(n-k)! + \binom{2n-k-1}{k-1}(n-k)! \\ &= \frac{2n(2n-k-1)!(n-k)!}{k!(2n-2k)!} \quad (\text{Work it out!}) \end{aligned}$$

Substituting this in our expression for S , we find that there are exactly

$$n! + \sum_{k=1}^n (-1)^k \frac{2n(2n-k-1)!(n-k)!}{k!(2n-2k)!}$$

ways for the n Knights to sit apart from their Ladies. Since there are a total of $n!$ possible seating arrangements for the n Knights, the probability that no couple sits together is

$$1 + \frac{1}{n!} \sum_{k=1}^n (-1)^k \frac{2n(2n-k-1)!(n-k)!}{k!(2n-2k)!},$$

or equivalently,

$$\frac{2}{(n-1)!} \sum_{k=0}^n (-1)^k \frac{(2n-k-1)!(n-k)!}{k!(2n-2k)!}.$$

By writing this expression in the form

$$1 + \sum_{k=1}^{n-1} \frac{(-2 + \frac{0}{n-k})(-2 + \frac{1}{n-k+1})(-2 + \frac{2}{n-k+2}) \cdots (-2 + \frac{k-1}{n-1})}{k!} + (-1)^n \frac{2}{n!},$$

the industrious reader can show that as the number of Knights and Ladies increases, the probability that no couple sits together approaches

$$1 + \sum_{k=1}^{\infty} \frac{(-2)^k}{k!} = e^{-2} \quad \text{as a limit.}$$

cont. from page 15

at a variation of a famous problem, and seeing what you can prove about it. Hopefully, the problem will intrigue you enough to get you to work on it night and day. The project may become more important than school, friends and family; in serious cases, it may override eating, sleeping, and the basic necessities of life. If your work has been going nowhere for a while, try not to despair, and branch out into new directions that may lead somewhere. Every so often, try to take your mind off your work.

Although invariably your work will start producing the best results just as the deadline is coming, remember that presenting your ideas in a logical, legible, and coherent fashion counts as well. Write a rough draft, look it over (if possible have a very experienced person review it), and make corrections. If you can finish the paper in advance, you won't have to pay the large fee needed to rush it down to Washington to meet the deadline. Don't underestimate the importance of the application form, on which you must give teacher-approved evidence of your scientific qualities. Start planning in advance to say brilliant things in the classroom and make certain that the teachers remember them. Most teachers will be very happy to help you fill out the forms, and will gain tremendous respect for you, which can't hurt at report card time.

Should you be successful/lucky enough to be in the winners group, you will experience something new. For the duration of a week, you will be the honored guest of a major corporation. You will have had arrangements made for you to see many sites in the city, including some which are not open to the general public in Washington. The company will pay for virtually all your claimed expenses, and will help you enjoy everything but the interviews. The interviewers will grill you, firing problems until they have made you succumb. They are prepared with guaranteed stumbers to see how you react to questions that you can't answer. After the judging was over, the judges revealed that they wanted to see if the subjects would know how to start work on a problem; they did not want the response they usually got, "I don't know."

Studying for an interview will probably have nothing better than a psychological effect, although attitude may be as crucial as anything else when being cross-examined. The judges later do apologize for knowing no better way to gauge your life potential in fifteen minutes.

In the time surrounding the trip, you will be celebrated by the media, getting more attention than you would ever anticipate would be given to a student. Colleges across the country should start sending you pantloads of mail, telling you that your award indicates your desire to attend a small liberal arts college.

For these reasons, it can be said that writing a nice paper can yield nice results, and it is probably worth the chance.

REGENTS TIPS

GEOMETRY

Supplementary angles add up to 180°

Complementary angles add up to 90°

Opposite angles of a quadrilateral in a circle add up to 180°

An equilateral triangle of side s has:

Area = $s^2\sqrt{3}/4$, Perimeter = $3s$, Altitude = $s\sqrt{3}/2$

Sum of angles in an n sided polygon

Interior angles: $180^\circ(n-2)$

Exterior angles: 360°

Tangents drawn from a point to a circle are equal in length and perpendicular to the radii drawn to the points of tangency.

Angles in circles:

The measure of an inscribed angle is $\frac{1}{2}$ the measure of the arc that it intercepts.

The measure of the angle between two secants is equal to $\frac{1}{2}$ the difference of the measure of the intercepted arcs.

Pythagorean Theorem: If a and b are the two legs of a right triangle, and c is the hypotenuse, then $a^2 + b^2 = c^2$

ALGEBRA & TRIG

The equation of a line is $y = mx + b$, where m is the slope, and b is the y -intercept.

x coordinate- abscissa

y coordinate- ordinate

tangent = $\frac{\text{opposite side}}{\text{adjacent side}}$

sine = $\frac{\text{opposite side}}{\text{hypotenuse}}$

cosine = $\frac{\text{adjacent side}}{\text{hypotenuse}}$

Law of cosines: $c^2 = a^2 + b^2 - 2ab \cos C$

Law of sines: $a/\sin A = b/\sin B = c/\sin C = 2R$ (R =circumradius)

$\cos(A+B) = \cos A \cos B - \sin A \sin B$

$\cos(A-B) = \cos A \cos B + \sin A \sin B$

$\sin(A+B) = \sin A \cos B + \cos A \sin B$

$\sin(A-B) = \sin A \cos B - \cos A \sin B$

Area of a Parallelogram: $K = ab \sin C$

If the amplitude of a sinusoidal curve is a and the frequency is b ,
period = $\frac{360}{b}$ degrees or $\frac{2\pi}{b}$ radians

equation of curve: $y = a \sin(bx)$

Math Team

Math team is an optional 0 period class which meets every morning in room 517. There, students are supposed to work on math problems and share their methods of solving them. Certain theorems, techniques, and illegal tricks are taught which help students get the right answers. The class is supervised by the lovable coach of the math team, our very own Dean Barry (Glotzer). Senior team captains Lisa Randall and Paul Feldman try to run the class, but it is the students who make the class interesting by demonstrating solutions and other assorted talents at the blackboard. Besides learning mathematics that might not be found in traditional classrooms, students are trained for various competitions on the local, state, and national level.

In competitions against other high schools in the city, the senior team won easily. The senior team consisted of Noam Elkies, Paul Feldman, Zac Franco, Steve Lev, Lisa Randall, and alternates Jae Chung, Gideon Eisenstadter, and Howard Reubenstein; another victory is imminent as the spring term competition draws to a close. Due to an overabundance of qualified students, especially juniors, two junior teams were allowed to compete for Stuyvesant. Unfortunately, our team "Alpha" only finished in second place. Fortunately, our team "Aleph" came in first place. Team members and alternates Patrick Davin, Leonid Fridman, Danny Glasser, Nathan Glasser, Joel Hirschhorn, Alan Horowitz, Fareed Khan, Calvin Lin, Gregg Patruno, Cathy Savage, Doris Stoffers, Ann Trenk, Joel Wein, and David Zagorski are expected to repeat last term's performance.

In the New York Math League competitions, Stuyvesant won first place with perfect scores on every meet; Noam Elkies had a perfect individual score. All students, not just team members, are invited to participate in these meets.

The Annual High School Mathematics Exam is also open to everyone. This year, the MAA, as it is more often referred to, was given on March 4. Gregg Patruno(145), Paul Feldman(145), and Zac Franco(135) had the highest combined team score in the nation. Other students who did well and qualified for National Honor Roll as well as the U.S. Olympiad include Noam Elkies(129), Steve Lev(125), Leonid Fridman(125), Lisa Randall(109), and Joel Hirschhorn(107).

Remember, all students are always welcome to attend math team, and it is a good idea to try it so you can take advantage of one of the things that makes our school the best in the country.

Exponential Population Growth

by Steve Lev

Exponential growth occurs when a fixed quantity increases by a constant percentage of itself in a given period of time. The human population is currently increasing at a phenomenal rate, and if present trends continue, will double in the short span of 33 years. This doubling time is roughly found by dividing 70 by the growth rate, say $r\% = \frac{r}{100}$. This is easily verified by considering the definition of exponential growth, $\frac{du}{dt} = \frac{r}{100} u$, where t is time. Separating variables and integrating both sides yields $\ln u + C = \frac{rt}{100}$. Since at time $t = 0$, we find that $C = -\ln u_0$, we can rewrite the preceding expression as $u = u_0 e^{rt/100}$. When the doubling time, t , has passed, $u_0 e^{r(t+\Delta t)/100} = 2u$, or after simplifying, $2 = e^{r\Delta t/100}$ solving for Δt , we finally arrive at $t = 100 \ln 2 / r \approx 69.3 / r \approx 70 / r$.

What is the reason for this recent rapid growth? The positive feedback loop involved with population is births per year, and the negative feedback loop is deaths per year. The rate of growth, is the difference between the two. In earlier times, the high fertility rate of mankind was almost balanced by a high mortality rate. Recently, however, the death rate has been falling off at a far quicker pace than the birth rate, due to cheap medical technology.

Several mathematical models have been proposed, attempting to explain the explosive population growth rate, the effects of which continually elude us.

One such model, named the Leslie Matrix after its discoverer, deals with the age distribution of a population during equal intervals of time. The conditions imposed on this analysis are that we only consider the females and that an age limit, L , exists.

It is thus possible to segment the population into the n age groups, $[0, \frac{L}{n}]$, $[\frac{L}{n}, \frac{2L}{n}]$, ..., $[\frac{(n-1)L}{n}, L]$. Let $x_y(0)$ = the number of females in the group $[\frac{(y-1)L}{n}, \frac{yL}{n}]$. Then the initial population distribution, x , may be written in matrix form as $\begin{bmatrix} x_1(0) \\ x_2(0) \\ \vdots \\ x_n(0) \end{bmatrix}$. Also, let the fraction of group j surviving at end of time t be b_j . Let the population distribution at time $t = \frac{L}{n}$ units later be $x^{(1)}$ and the average number of daughters

for each member of group j be a_j .

It then follows that

$$\begin{aligned} x_1^{(1)} &= a_1 x_1^{(0)} + a_2 x_2^{(0)} + \dots + a_n x_n^{(0)} \\ x_{y+1}^{(1)} &= b_y x_y^{(0)} \end{aligned}$$

We can simplify these equations by creating the Leslie Matrix, L , equal to

$$L = \begin{vmatrix} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & 0 & 0 & \dots & 0 \\ 0 & b_2 & \dots & \dots & 0 \\ 0 & \dots & \dots & \dots & b_{n-1} \end{vmatrix}$$

Then $x^{(1)} = L x^{(0)}$, $x^{(2)} = L x^{(1)} = L^2 x^{(0)}$, ... and $x^{(n)} = L^n x^{(0)}$.

Assuming the age distributions are constant with respect to time, then there must be some scalar for which $Lx = \lambda x$, where the real number λ is known as an eigenvalue and x is called an eigenvector.

Simplifying, we get $Lx - \lambda x = (L - \lambda I)x = 0$, I the identity matrix.

$$\lambda I - L = \begin{vmatrix} -a_1 & -a_2 & -a_3 & \dots & -a_n \\ -b_1 & & 0 & \dots & 0 \\ 0 & -b_2 & & \dots & 0 \\ 0 & 0 & & \dots & -b_{n-1} \end{vmatrix}$$

and the determinant of $(\lambda I - L) = \lambda^n - a_1 \lambda^{n-1} - a_2 b_1 \lambda^{n-2} - \dots - a_n b_1 b_2 \dots b_{n-1}$.

This expression is called the characteristic polynomial of L , and has a unique positive eigenvalue λ_1 . Now let x^0 be any starting population.

Then $x^0 = w_1 x_1 + w_2 x_2 + \dots + w_n x_n$ and $L^y x^0 = w_1 \lambda_1^y + w_2 \lambda_2^y x_2 + \dots + w_n \lambda_n^y x_n$.

Thus, $[L^y x^0] [\frac{1}{\lambda_1^y}] = w_1 x_1 + w_2 [\frac{\lambda_2}{\lambda_1}]^y x_2 + \dots + w_n [\frac{\lambda_n}{\lambda_1}]^y x_n$.

As y approaches infinity, $\frac{1}{\lambda_1^y} L^y x^0 = w_1 x_1$. This is the desired result, that $L^y x^0$, or $x^{(1)} = \lambda_1^y (w_1 x_1)$, clearly an exponential expression.

The fundamental problem of continued exponential growth is that food and resources are limited. It has been projected that, if present growth trends continue, there will be a drastic shortage of arable land before the year 2000. However, given only a rational control of population, the prospects of future economic and social growth are exceedingly high.

Fermat's Last Theorem

by Joel Wein

For hundreds of years, mathematicians have been unable to prove Fermat's Last Theorem. When Fermat scribbled the theorem in the margin of a book, he claimed: "I have a marvelous demonstration of this, but the margin is too narrow to contain it." Whether or not he had a real proof is questionable, but since then no one has been able to prove the entire theorem. Many special cases have been proven, and I will demonstrate one here.

Fermat's Last Theorem states that for c, d, e positive integers and n an integer > 2 , $c^n + d^n = e^n$ has no solutions. (For $n = 2$ we have our old friends the Pythagorean triplets: $(3, 4, 5)$, $(8, 15, 17)$, etc.) Here we will prove the theorem for $n=4$. In fact, a stronger result will be proven: that $c^4 + d^4 = e^2$ has no solutions for positive integers c, d, e .

Assume there are solutions to $c^4 + d^4 = e^2$. One of them must give us the smallest value of e for all such triplets. Let this triplet be (x, y, z) . It is easy to show that x, y , and z share no common factors (are relatively prime).

Since $(x^2)^2 + (y^2)^2 = z^2$, (x^2, y^2, z) is a Pythagorean triplet. There is a theorem in Number Theory which states that for any Pythagorean triplet (s, t, q) where s, t , and q are relatively prime, there exist relatively prime integers a and b such that $s=2ab, t=a^2-b^2$, and $q=a^2+b^2$. Letting $x^2=s, y^2=t$, and $z=q$, we find that we can express $x^2=2ab, y^2=a^2-b^2$, and $z=a^2+b^2$.
 $y^2 = a^2 - b^2$ $b^2 + y^2 = a^2$, so (b, y, a) is a Pythagorean triplet.

As was stated above, we can now find two relatively prime integers k and m such that

$$b=2km, y=k^2-m^2, a=k^2+m^2.$$

Since $2ab=x^2$, and a and b are relatively prime, either a is a square or b is a square. Without loss of generality, assume that $a = r^2$, so b must be twice a square. Let $b = 2s^2$. $b = 2s^2 = 2km$, so $s^2=km$. Since k and m are relatively prime, k must be a square, say $k = u^2$, and $m = v^2$. Then $k^2+m^2 = a = r^2$, and $u^4+v^4=r^2$, since $k=u^2$, and $m=v^2$. Since $z^2=a^2+b^2$, and $a = r^2$, $z > r$. Thus we have found a new triplet (u, v, r) which has a smaller value of e (namely r). But we assumed that z was the smallest value of e . This assumption leads to a contradiction (that there was a smaller e), which means that we can never find a smallest solution $c^4+d^4=e^2$. Therefore we can never find any solutions, and Fermat's Last Theorem is true in the special case $n = 4$.

Game Theory

by Joel Hirschhorn

Game theory was introduced in a book called The Theory of Games and Economic Behavior, by John von Neumann and Oskar Morgenstern. Not much attention was paid to game theory until it was used successfully by the Allies in World War II to minimize the loss of ships due to U-boats. Game theory then became an important topic in mathematics.

Game theory is the analysis of a "game"; that is, determining the best strategy for each player and the result, assuming each player uses this strategy. This article will examine games where two players, X and Y, compete.

Before continuing, a few definitions are necessary.

Payoff: how much one player must pay another if they choose certain strategies.

Oddments: the ratios of the number of times a player should play his strategies (e.g. if a player's oddments are 3:2:0, he should play strategy #1 $\frac{3}{5}$ of the time, strategy #2 $\frac{2}{5}$ of the time, and never play strategy #3).

Value of the game: how much one player must pay the other on the average if both play ideally.

A game is solved if the oddments for both players and the value of the game are found.

A two person game is symbolized by a $m \times n$ matrix, where player X has n strategies and player Y has m . The k th strategy of player X is called X_k . The payoff from Y to X if strategies X_i and Y_j are played is placed in the i th row and the j th column.

	Y1	Y2	Y3
X1	5	0	1
X2	-4	2	0

In this example, if strategies X1 and Y3 are played, Y pays X 1 unit; if strategies X2 and Y1 are played, X pays Y: 4 units.

There are various methods of solving "games". One is the minimax method. It is as follows:

1) Find the minimum payoff in each row, and take the maximum of these minimums (the maximin).

2) Find the maximum payoff in each column, and take the minimum of these maximums (the minimax). There is a theorem which states that if the minimax equals the maximin, then a "saddle point" exists, and X and Y should use the strategies corresponding to the maximin and the minimax, respectively. In addition, the value of the game is equal to the minimax.

Dominance

If one strategy is always worse than another one, it should not be used. This is intuitively obvious. Eliminating the worse, or dominated strategy invariably makes the game easier to solve.

Oddments

In the simplest case, a 2×2 game, it may be shown that for the game

Figure (1)

	Y1	Y2
X1	C1	C2
X2	C3	C4

where C1, C2, C3 and C4 are payoffs, X's oddments are $C4-C3:C1-C2$ and Y's are $C4-C2:C1-C3$.

Once the oddments are known, the value of the game can be found. To find it, use the following method:

Take the probability of a certain payoff occurring, and multiply it by the value of the payoff. The probability of the payoff resulting from Xi vs. Yj occurring is the probability of X choosing Xi times the probability of Y choosing Yj. The probability of X choosing Xi is Xi's oddment divided by the sum of X's oddments (e.g. if X's oddments are 3:4, the probability of him choosing X2 is $4/(3+4)=4/7$). Similarly, the probability of Y choosing Yj is Yj's oddment divided by the sum of Y's oddments. Therefore, the probability of the occurrence of the payoff is:

$$\frac{(\text{Xi's oddment})(\text{Yj's oddment})}{(\text{sum of X's oddments})(\text{sum of Y's oddments})}$$

When this is multiplied by the payoff, let the result be p. Repeat the process for all of the payoffs, and add up all the p's to obtain the value of the game.

Example: (X's oddments are on the right, and Y's are on the bottom of the matrix.)

Figure (2)

	Y1	Y2	
X1	8	2	4
X2	3	7	6
	5	5	

$$\text{Value} = \frac{(4)(5)(8)}{(4+6)(5+5)} + \frac{(4)(5)(2)}{(4+6)(5+5)} + \frac{(6)(5)(3)}{(4+6)(5+5)} + \frac{(6)(5)(7)}{(4+6)(5+5)} = 5$$

Referring to the matrix in figure (1), if X uses the oddments $C4-C3:C1-C2$, he will win the same amount no matter how Y plays. To show this, we let Y's oddments be $y:n-y$, where n is any number. Then,

$$\begin{aligned} \text{Value} = & \frac{(C4-C3)(y)(C1)}{(C4-C3+C1-C2)(n)} + \frac{(C4-C3)(n-y)(C2)}{(C4-C3+C1-C2)(n)} + \frac{(C1-C2)(y)(C3)}{(C4-C3+C1-C2)(n)} + \\ & + \frac{(C1-C2)(n-y)(C4)}{(C4-C3+C1-C2)(n)} = \frac{(C1)(C4) - (C2)(C3)}{(C4-C3+C1-C2)} \end{aligned}$$

Formula (1)

The Brachistochrone Problem

by Nathan Glasser

A notable problem in physics as well as mathematics is the Brachistochrone problem, which consists of the following: we are given two points in space, and we want to find the path a point-mass should follow in order to reach the second point in the least amount of time under uniform acceleration due to gravity. Construct a vertical coordinate plane through the two points so that the uppermost point is at the origin and the fixed point at (x_0, y_0) , and take the positive y direction to be downward. We will now try to find a function $y = y(x)$ passing through $(0, 0)$ and (x_0, y_0) whose graph is the desired curve. Let m be the mass of the moving particle. Then while at any point (x, y) , its velocity v will satisfy $\frac{1}{2}mv^2 = mgy$ because the kinetic energy equals the difference in potential energy. Upon rearranging, $v = \sqrt{2gy}$. Since $\bar{v} = ds/dt$, the time required by the particle to travel along an element of arc length ds of the curve is given by

$$ds/v = \frac{\sqrt{1+(y')^2}}{\sqrt{2gy}} dx$$

Integrating, the total traveling time along the curve equals

$$\frac{1}{\sqrt{2g}} \int_0^{x_0} \frac{\sqrt{1+(y')^2}}{y} dx$$

We must now find the function y which minimizes this integral function of y . To this end we employ the calculus of variations. Euler's equation states that in order for a function $y(x)$ to minimize the functional

$$I(y) = \int_{x_1}^{x_2} F(x, y, y') dx,$$

x and y must satisfy the differential equation $F_y = d/dx(F_{y'})$, where the subscripts denote partial derivatives.

In this problem,

$$F = \frac{\sqrt{1+(y')^2}}{y}, \text{ so } F_y = -\frac{\sqrt{1+(y')^2}}{y^2}, \text{ } F_{y'} = \frac{y'}{y\sqrt{1+(y')^2}}$$

$$\begin{aligned} \text{and } d/dx(F_{y'}) &= (F_{y'})_{y'} y' + (F_{y'})_{y''} y'' \\ &= \frac{y''}{\sqrt{y} \sqrt{1+(y')^2}} - \frac{(y')^2}{2y^{3/2} \sqrt{1+(y')^2}} = \frac{(y')^2 y''}{\sqrt{y} (1+(y')^2)^{3/2}} \end{aligned}$$

$$= -\frac{\sqrt{1+(y')^2}}{2y^{3/2}} \quad \text{by Euler's equation. Simplifying,}$$

$$-\frac{1}{y} = \frac{2y''}{1+(y')^2}.$$

Multiplying by dy and integrating gives $-\ln(y) = \ln(1+(y')^2) + A$. Hence $B/y = 1+(y')^2$, and $\pm dy \sqrt{y/B-y} = dx$. Making the substitution $y = \frac{1}{2}B(1-\cos u)$, $dy = \frac{1}{2}B \sin u \, du$ reduces the differential equation to $dx = \pm \frac{1}{2}B(1-\cos u)du$, $x = \pm \frac{1}{2}B(u - \cos u) + C$. Using the fact that $x \geq 0$ and that the curve passes through $(0,0)$ gives $x = \frac{1}{2}B(u - \sin u)$. From this we can see that the desired curve is a cycloid. (These are the parametric equations of a cycloid.) It can be proven that the lowest point of the arch of the cycloid passes through (x_0, y_0) , and thus the constant $\frac{1}{2}B$ can be determined, giving the particular solution. An interesting fact about the cycloid is that if a point is released anywhere on the curve between the two given, the time it will take to fall to the fixed point is the same as the time if released at any other point on the curve.

For more on Euler's equation and the calculus of variations, see:

Mathematics: Its content, Methods, and Meaning, Vol II
by A. D. Aleksandrov

Calculus of Variations
by G. A. Bliss

PROBLEMS

1. Solve for all real x :

$$\sqrt{\frac{3x+2}{6x-4}} + \sqrt{\frac{6x-4}{3x+2}} = \frac{5}{2}.$$

2. In $\triangle ABC$, $AB = 13$, $BC = 14$, and $CA = 15$. A line parallel to BC intersects AB at D and AC at E . If $BD + EC = \frac{1}{2}BC$, find the distance from A to DE .

3. For what value of x does the continued fraction

$$x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$$

have the value 3, if the dots indicate an infinite repetition of the indicated pattern?

Rabbits and Numbers



by Doris Stoffers and Ann Trenk

In the 12th century A.D., Leonardo Fibonacci wrote a famous book on arithmetic called the Liber Abaci. In it he proposed a mathematical puzzle concerning the breeding of rabbits. It went as follows: beginning with one rabbit, if this rabbit produced another rabbit every month, and if every newborn rabbit began reproducing at the rate of one rabbit per month two months after its birth, how many rabbits would there be at the end of n generations? The solution of this problem led to the development of the sequence:

1, 1, 2, 3, 5, 8, 13, ...

Each term (except the first two) is the sum of the two terms before it, the n^{th} term represents the total number of rabbits after n months. The sequence is called the Fibonacci sequence, and its terms are called Fibonacci numbers. In symbols, the recursion formula states:

$$F_1 = F_2 = 1; \quad F_n = F_{n-2} + F_{n-1} \quad (\text{where } n > 2 \text{ and } F_n \text{ is the } n^{\text{th}} \text{ Fibonacci number})$$

Another interesting property of this sequence is that the sum of the first n terms is equal to the $(n+2)^{\text{nd}}$ term minus one. This can be proven by mathematical induction.

Using the information we have been given up to this point, it is possible to find the n^{th} term of the Fibonacci sequence only by first finding the preceding $n-1$ terms. However this is very inconvenient and so, we would like to find a general formula for the n^{th} Fibonacci number.

Denote the n^{th} Fibonacci number by a_n . Consider the function $f(x)$ where x^k has the coefficient a_k . Then,

$$f(x) = 1 + x + 2x^2 + 3x^3 + 5x^4 + \dots + a_n x^{n-1} + \dots$$

and, $f(x) - 1 = x + 2x^2 + 3x^3 + 5x^4 + \dots + a_n x^{n-1} + \dots$
Also, we know that

$$xf(x) = x + x^2 + 2x^3 + 3x^4 + \dots + a_{n-1} x^{n-1} + \dots$$

and, $x^2 f(x) = x^2 + x^3 + 2x^4 + \dots + a_{n-2}x^{n-1} + \dots$
 We observe that $f(x) - 1 = xf(x) + x^2 f(x)$
 Solving for $f(x)$ yields

$$f(x) = \frac{-1}{x^2 + x - 1}$$

We know that we may factor $x^2 + x - 1$ into $(x - r_1)(x - r_2)$, where r_1 and r_2 are the roots of the equation $x^2 + x - 1 = 0$. Using the quadratic formula we find

$$r_1 = \frac{-1 + \sqrt{5}}{2} \quad \text{and} \quad r_2 = \frac{-1 - \sqrt{5}}{2}$$

We may now say that

$$f(x) = \frac{-1}{(x - r_1)(x - r_2)}. \quad \text{Factoring out } \frac{-1}{(r_1 - r_2)} \text{ we obtain}$$

$$f(x) = \left(\frac{-1}{r_1 - r_2} \right) \left(\frac{r_1 - r_2}{(x - r_1)(x - r_2)} \right)$$

We know $\frac{r_1 - r_2}{(x - r_1)(x - r_2)} = \frac{1}{x - r_1} - \frac{1}{x - r_2}$ so that substituting, we find

$$\begin{aligned} f(x) &= \left(\frac{-1}{r_1 - r_2} \right) \left(\frac{1}{x - r_1} - \frac{1}{x - r_2} \right) \\ &= \left(\frac{1}{r_1 - r_2} \right) \left(\frac{1}{r_1 - x} - \frac{1}{r_2 - x} \right) \\ &= \frac{1}{r_1 - r_2} \left(\frac{1}{r_1} \cdot \frac{1}{1 - \frac{x}{r_1}} - \frac{1}{r_2} \cdot \frac{1}{1 - \frac{x}{r_2}} \right) \end{aligned}$$

It's now consider the infinite geometric $1, a, a^2, \dots$ where $|a| < 1$. Then a sum of this series is $\frac{1}{1 - a}$.

To apply this to our proof we will make x sufficiently small such that, $|x/r_1| < 1$ and $|x/r_2| < 1$. Substituting x/r_1 and x/r_2 for " a ", yields

$$\frac{1}{1 - \frac{x}{r_1}} = 1 + \frac{x}{r_1} + \frac{x^2}{r_1^2} + \dots \quad \frac{1}{1 - \frac{x}{r_2}} = 1 + \frac{x}{r_2} + \frac{x^2}{r_2^2} + \dots \quad \text{so that}$$

$$f(x) = \frac{1}{r_1 - r_2} \left[\frac{1}{r_1} \left(1 + \frac{x}{r_1} + \frac{x^2}{r_1^2} + \dots \right) - \frac{1}{r_2} \left(1 + \frac{x}{r_2} + \frac{x^2}{r_2^2} + \dots \right) \right]$$

$$f(x) = \frac{1}{r_1 - r_2} \left[\left(\frac{1}{r_1} + \frac{x}{r_1^2} + \frac{x^2}{r_1^3} + \dots \right) - \left(\frac{1}{r_2} + \frac{x}{r_2^2} + \frac{x^2}{r_2^3} + \dots \right) \right]$$

We can now find the coefficient of x^n , which is

$$a_n = \left(\frac{1}{r_1 - r_2} \right) \left(\frac{1}{r_1^{n+1}} - \frac{1}{r_2^{n+1}} \right)$$

Substituting the values of r_1 and r_2 found earlier gives the formula for the n^{th} Fibonacci number,

$$a_n = \left[\frac{1}{\sqrt{5}} \left(\frac{2}{-1+\sqrt{5}} \right)^{n+1} - \left(\frac{2}{-1-\sqrt{5}} \right)^{n+1} \right]$$

In deriving this formula, we have come across the number, $(1+\sqrt{5})/2$, which is known as the Golden Ratio. When the ratio of length to width of a rectangle equals the Golden Ratio, it is known as a Golden Rectangle. Two German psychologists have shown in a series of experiments that most people unconsciously favor objects with golden dimensions when choosing pictures, cards, mirrors and other rectangular objects. For some reason not known, the Golden Rectangle holds great aesthetic appeal.

When we take ratios of consecutive Fibonacci numbers we notice that as the Fibonacci numbers increase their ratios approach the Golden Ratio. Consequently, the best integral lengths to use for approximations of the Golden Rectangle are consecutive terms of the Fibonacci sequence. For example, we have 3x5 and 5x8 index cards.

The Fibonacci numbers have uses in seemingly unrelated fields, such as electrical network theory, phyllotaxis, genealogy, and architecture. The sequence is being actively researched in the hope of finding more properties and practical applications.

THE YELLOW PIG EXPANSION

$$\left(\begin{array}{c} \text{yellow} \\ \text{can} \end{array} + \begin{array}{c} \text{pig} \\ 16 \end{array} \right)^2 = \begin{array}{c} \text{yellow} \\ \text{square} \end{array} + \begin{array}{c} \text{yellow} \\ \text{pig} \\ 17 \end{array} + \begin{array}{c} \text{yellow} \\ \text{pig} \\ 17 \end{array} + \begin{array}{c} \text{square} \\ \text{pig} \end{array}$$

cont. from page 7

$$\prod_{i=1}^{\infty} (1 + \frac{1}{p_i} + \frac{1}{p_i^2} + \dots) = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

This sum is infinite (see any calculus text for a proof), and therefore so is $\prod_{i=1}^{\infty} \frac{1}{1 - (1/p_i)}$.

But if there were a finite number of primes, the product would be finite. Therefore there are infinitely many primes.

Proof 4: This is the classic proof of Euclid. Assume there are finitely many primes; call them $p_1, p_2, p_3, \dots, p_n$. Consider the number $N = (p_1 p_2 p_3 \dots p_n) + 1$. It is not divisible by any of the n primes listed above (N leaves a remainder of 1 when divided by each of these primes). But every integer greater than 1 has at least one prime factor, so there must be a prime that is not included in our list. This contradicts our assumption that we could find a complete list of primes. Therefore, there are infinitely many primes.

A Mathematician's Bookshelf

1. A Short Table of Even Primes (abridged)
2. The Peano Postulates Transcribed for Violin and Cello
3. History of Arcs, by Joan
4. 1,000,000 Random Numbers in Ascending Order
5. Integration in Closed Neighborhoods
6. Homemade Rolles, Mersenne Prime Ribs, Cauchy Dill Pickles, Transcendental Pi and Other Tasty Treats
7. 22/7 Computed to One Million Decimal Places
8. The Jacobians and Their Struggle for Independence
9. Tom Swift and His Electric Cycloid
10. Logarithm Tables and Other Antique Furniture
11. Eccentricities of Orbiting Mathematicians
12. First Aid for Dedekind Cuts
13. The Difference Between a Doughnut and a Coffee Cup- a Guide to Practical Living for Topologists
14. Rings and Ideals- a Modern Marriage Manual
15. Transfinite Counting for Insomniacs
16. A Complete Collection of Magic Squares of Order Two
17. L'Hôpital's Diet for Improving Indeterminate Forms

Since the value of the game is not affected by y or n , if X plays ideally, Y can not change the value of the game.

To illustrate the application of Formula (1), we use the matrix in Figure (2).

$$\text{Value} = \frac{(8)(7) - (2)(3)}{(7-3+8-2)} = 5$$

As can be seen, Formula (1) provides a much simpler way of finding the value of a 2×2 game as compared to the method described previously.

2 x m games

These games are slightly harder to solve than 2×2 games.

The method for solving them is as follows:

1) Search for a "saddle point". If one exists, then solve the game using the minimax method.

2) Eliminate all dominated strategies. If a 2×2 game results, solve it as described previously.

3) If neither of these two methods apply, then find the value of every 2×2 game which is included in the $2 \times m$ game. Choose the game which gives the best result for the player with m strategies, and solve it (i.e. if X has m strategies, choose the game with the highest value; if Y has m , then choose the game with the lowest value). The value of this game is equal to the value of the $2 \times m$ game, and the oddments of the 2 strategies contained in the 2×2 game are the oddments of these strategies in the large game. The oddments for the rest of the strategies of the player who has m strategies are all 0. The oddments of the strategies of the player who has 2 strategies are the same as they are in the 2×2 game.

Example:

	Y1	Y2
X1	5	1
X2	4	3
X3	2	5

There is no "saddle point" or dominance, so, since X has m strategies, we must choose the 2×2 game which has the highest value.

The value of the game containing X1 and X2 is (by minimax) 3.

The value of the game containing X1 and X3 is (by Formula 1)

$$\frac{(5)(5) - (1)(2)}{(5-2+5-1)} = \frac{23}{7}.$$

The value of the game containing X2 and X3 is (by Formula 1)

$$\frac{(4)(5) - (3)(2)}{(5-2+4-3)} = \frac{7}{2}.$$

Since the game containing X2 and X3 has the highest value, we solve this game. Using the formula shown under Figure 1, we find

X's oddments to be 3:1, and Y's to be 2:2. Since strategy X1 was not included in the 2×2 game, its oddment is 0. Also, the value of the 2×2 game was $7/2$, so the value of the 2×3 game is $7/2$. The solution is: X's oddments are 0:3:1, Y's are 2:2, and the value of the game is $7/2$.

This article has examined methods of solving 2×2 and $2 \times m$ games. A method of approximating the solution to more complex games may be found in The Theory of Games and Economic Behavior, third edition, 1953 by John von Neumann and Oskar Morgenstern or any other recent text on the subject.

Game theory has many uses: in economics, warfare, politics and other fields. It is not, as the title seems to imply, only for leisure. Instead, game theory is a field of mathematics which has almost unlimited applications, and in which new discoveries are still being made.

CONTEST PROBLEMS

CONTEST PROBLEMS

This contest is open to all students of Stuyvesant High School except for those associated with the Math Team. Solutions should be written on separate sheets of paper, and will be judged on correctness, completeness, and clarity. Awards will be presented for the best solutions at the Math Department Luncheon (to be held in June). Solutions must be brought to Room 517 before 1st Period or submitted to Mr. Marcantonio no later than May 30, 1980.

Good luck!

1. Prove that in the infinite sequence of integers
10001, 100010001, 1000100010001, ...
there is no prime number.
2. At midnight a truck starts from city A and goes to city B. At 2:40 a.m. a car starts from B to A along the same route. They pass at 4 a.m. The car arrives at its destination 40 minutes later than the truck. Eventually, they start back and pass each other at 2 p.m. They arrive at the same time. Find it.
3. A "self-avoiding walk" is a sequence of steps of length one taken in directions North, South, East, or West that never passes through the same point twice. Let $F(n)$ denote the number of n -step self-avoiding walks which begin at the origin. Compute $F(1)$, $F(2)$, $F(3)$, $F(4)$ and show that
$$2^{n+2} - 4 \leq F(n) \leq 4 \cdot 3^{n-1}.$$
4. Prove that the fraction $\frac{34n+9}{51n+13}$ is irreducible.

POISSON SEXTUPLETS

by Paul Feldman

What is the probability that there will be exactly 3 multiple births in the coming year involving at least 6 children each? This question could be quite hard to answer unless some information is given.

Assume that the chance of a birth being a multiple birth remains constant, and that each year the same number of births occur. In the last 80 years, 50 sets of 6 or more children have been born, an average of .625/year. This may not seem like enough information, but it actually is sufficient to get an excellent approximation.

If N is the number of births, and p is the chance of a birth involving 6 or more children, the exact probability =

$\binom{N}{3} \cdot p^3 \cdot (1-p)^{N-3}$ (The number of ways to select 3 births out of N is $\binom{N}{3}$, the chance that each is a multiple birth is p^3 , and the chance that none of the others are is $(1-p)^{N-3}$.)

We cannot evaluate the probability exactly, since the only equation we have is $p \cdot N = .625$, the expectation (average). However, if we can assume that N is very large and p is very small (which seem to be valid assumptions in this example), we can approximate

$$\binom{N}{3} p^3 \cdot (1-p)^{N-3} \doteq \frac{N \cdot (N-1) \cdot (N-2)}{3 \cdot 2 \cdot 1} \cdot p^3 (1-p)^N \doteq \frac{N^3 \cdot p^3 \cdot (1-p)^N}{3!}$$

For convenience, let $\lambda = N \cdot p$ ($= .625$ in this case.) Using the fact that $\lim_{N \rightarrow \infty} (1 + \frac{x}{N})^N = e^x$,

$$\frac{N^3 \cdot p^3 \cdot (1-p)^N}{3!} \doteq \frac{(1 + \frac{-\lambda}{N})^N \cdot \lambda^3}{3!} \\ \doteq \frac{e^{-\lambda} \cdot \lambda^3}{3!}.$$

This approximation is the foundation of Poisson distribution, which can be used for various other problems. Incidentally, the approximation given is $\frac{(.625)^3 \cdot e^{-.625}}{3!}$

which is about equal to .02178.

In general, the probability of k occurrences will be $\binom{N}{k} p^k (1-p)^{N-k} \doteq \frac{N^k \cdot p^k}{k!} (1-p)^N \doteq \frac{e^{-\lambda} \cdot \lambda^k}{k!}$, where $\lambda = N \cdot p$.

To show that the sum of all the probabilities = 1, consider

$$\lim_{k \rightarrow \infty} \sum_{i=0}^k \frac{e^{-\lambda} \cdot \lambda^i}{i!} = e^{-\lambda} \cdot \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{-\lambda} \cdot e^{+\lambda} = 1.$$

Thus, as N grows very large, the approximation approaches the actual probability. Poisson distribution is useful when: hiring enough switchboard operators so they can just handle

the peak number of calls; finding the probability that milk from cows in Harrisburg, Pa. will have too much radioactivity to drink; and anticipating how many times you must ride before catching a relatively empty F train.

We might expect that the highest probability will occur at the integer nearest to the expected number of outcomes (λ), but that is not exactly true. To find the highest probability, consider the ratio of the probability of k outcomes to the probability of $k+1$ outcomes, =

$$\frac{\frac{e^{-\lambda} \cdot \lambda^{k+1}}{(k+1)!}}{\frac{e^{-\lambda} \cdot \lambda^k}{k!}} = \frac{\lambda}{1+k}, \text{ simply.}$$

From this it can be seen that the chance of $k+1$ occurrences is better than the chance of k occurrences if and only if $k+1 < \lambda$. Thus, the probability increases with increasing values of k until $k+1 > \lambda > k$, at which point the probability decreases with increasing values of k . Thus, the maximum is found when $k = [\lambda]$, where $[x]$ represents the greatest integer $\leq x$. In the case that λ is an integer, λ and $\lambda-1$ are tied for the most likely number of occurrences.

We can get a very good approximation to the chance that the expected number of occurrences will occur if it is fairly large. We can then use Stirling's Approximation, which is usually used for integers, but by considering the factorial function to be defined on non-integers to form a smooth curve, then for large λ , we can use the approximation, which says: $n! \approx n^n \cdot e^{-n} \cdot (2\pi n)^{1/2}$. Substituting this into the probability

of getting λ occurrences, $\frac{e^{-\lambda} \cdot \lambda^\lambda}{\lambda!} \approx \frac{e^{-\lambda} \cdot \lambda^\lambda}{\lambda^\lambda \cdot e^{-\lambda} \cdot (2\pi \lambda)^{1/2}} = \frac{1}{(2\pi \lambda)^{1/2}}$. Curiously, this is $1/2^{1/2}$ as large as the probability of getting the expected number of outcomes when $p=1/2$, which = $\binom{N}{\lambda/2} \cdot p^{(\lambda/2)} \cdot (1-p)^{(N-\lambda/2)} = \binom{N}{\lambda/2} \cdot (1/2)^N$ (using Stirling to estimate $\binom{2\lambda}{\lambda} = \frac{(2\lambda)!}{(\lambda!) \cdot (\lambda!)}$)

$$\frac{((2\lambda)^{2\lambda}) \cdot (e^{-2\lambda}) \cdot (4\pi \lambda)^{1/2}}{\lambda^{2\lambda} \cdot e^{-2\lambda} \cdot 2\pi \lambda} \cdot \frac{1}{2^{2\lambda}} = \frac{1}{(\pi \lambda)^{1/2}}.$$

One last interesting fact which is easily provable is that the probability of an even number of outcomes is always greater than the probability of an odd number of outcomes. You might want to try to reason this out logically; mathematically, the sum of the terms which correspond to an even number of occurrences minus the sum of the terms which correspond to an odd number of occurrences = $\sum_{k=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^{2k}}{(2k)!} - \sum_{k=0}^{\infty} \frac{e^{-\lambda} \cdot \lambda^{2k+1}}{(2k+1)!} =$

$$\sum_{k=0}^{\infty} \frac{e^{-\lambda} \cdot (-\lambda)^k}{k!} = e^{-\lambda} \cdot \sum_{k=0}^{\infty} \frac{(-\lambda)^k}{k!} = e^{-\lambda} \cdot e^{-\lambda} = e^{-2\lambda},$$

and this difference is always positive.

NEW YORK UNIVERSITY
DEPARTMENT OF MATHEMATICS

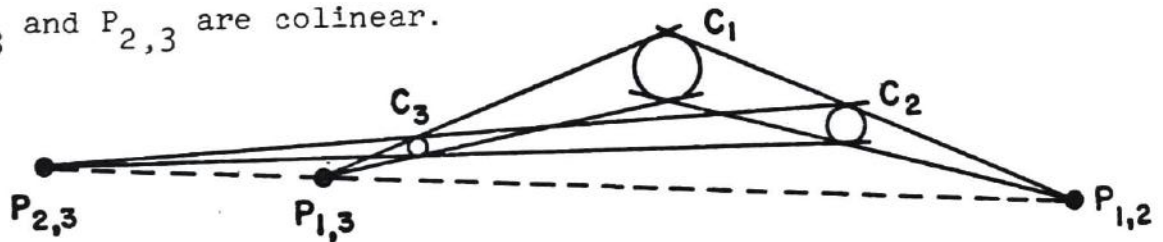
PRESENTS:

.....SOME BRAIN TEASERS TO TEST YOUR METTLE. A SPLENDID AWARD
WILL BE MADE FOR THE BEST SOLUTIONS. GOOD LUCK!

1. Let x, y satisfy $x^2 + y^2 = 3$
 $x^3 + y^3 = 4.$

What are the possible values of $x^4 + y^4$?

2. Let C_1, C_2, C_3 be three pairwise incongruent non-overlapping circles. Any two, C_i and C_j , have two tangent lines $\ell_{i,j}$ and $\ell'_{i,j}$ each of which is a line of support for both C_i and C_j . Let $\ell_{i,j}$ and $\ell'_{i,j}$ intersect at $P_{i,j}$ and show that $P_{1,2}, P_{1,3}$ and $P_{2,3}$ are collinear.



3. Show that if n_k is an increasing sequence of positive integers then $\sum_{k=1}^{\infty} \frac{1}{(n_k)!}$ is irrational.
4. Show that if $n + 1$ integers are chosen from $1, 2, \dots, 2n$ then one chosen number must divide another chosen number.

Send your solutions to: DEPARTMENT OF MATHEMATICS . C/O DOYLE
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DEADLINE: July 1, 1980